2023 3rd International Conference on Neural Networks, Information and Communication Engineering (NNICE) | 979-8-3503-3597-2/23/$31.00 ©2023 IEEE | DOI: 10.1109/NNICE58320.2023.10105766

MTL-JER: Meta-Transfer Learning for Low-Resource Joint Entity and Relation Extraction

Da Peng
Science and Technology on Complex Electronic System Simulation Laboratory
Space Engineering University
Beijing, China
wushanpengda@163.com

Zhongmin Pei*
Science and Technology on Complex Electronic System Simulation Laboratory
Space Engineering University
Beijing, China
*Corresponding author: peizhongmin@tsinghua.org.cn

Delin Mo
Institute of War Studies
Academy of Military Sciences
837613623@qq.com

***Abstract*—Joint entity and relation extraction has achieved impressive advances in NLP, such as document understanding and knowledge graph construction. The typical methods for entity and relation extraction typically break down the joint task into several smaller components or stages for ease of implementation, but this leads to a loss of the interconnected knowledge in the triple. Hence, we propose to model the triple in one module jointly. Furthermore, the labeling of a joint entity and relation extraction tasks is costly and domain-specific; therefore, it is important to improve its performance on low-resource data and domain adaption. To address this issue, we suggest using two sources that are rich in information, namely pre-trained models on large data and multi-domain text corpora. Pretraining allows us to provide the model with the fundamental ability to perform joint entity and relationship extraction. Second, through meta-learning on multi-domain text, we can improve the model's generalization capabilities, enabling it to perform well even with limited data. We present MTL-JER, a Meta-Transfer Learning method for Joint Entity and Relation Extraction in low-resource settings in this paper. Using exhaustive experiments on five datasets, we prove that our model obtains optimal results.**

Index Terms-Joint entity and relation extraction, Meta-Learning, Low-resource

I. INTRODUCTION

The aim of a joint entity and relation extraction is to extract pairs of entities and corresponding relations in the triple format (subject, relation, object) or (s,p,o) from unstructured text. It's a separate task for pipeline methods ([1, 2]) that named entity recognition and relation classification, which ignores the intersection between entity and relation, and also exist the error propagation problem [3]. Therefore, recent research focuses on modeling the entity and relation in one joint framework.

Existing methods for joint entity and relation extraction usually decompose this problem into multi modules or steps. Another work has proposed a unified sequence labeling framework by using a new decomposition method [4]. Casrel [5] uses BERT as a backbone model and passes the embedding of entity mentions into relations in a novel tagging framework, which takes multi-modules and multi-steps. Tplinker [6] formulates the joint extraction task as a token pair linking problem and uses a novel handshaking tagging paradigm that aligns the boundary tokens of entity pairs for each relation type, which takes multi-modules. The module of these models use the table-filling paradigm, which predicts the start and end index to extract entities. The model confirms an entity by predicting the position of its start and head tokens. For each relation, the tp-linker constructs two tables for subject and object prediction, which causes much redundancy and computation overload when the quantity increases. Furthermore, the decoding stage of tplinker is still in the pipeline paradigm, which exists in error propagation. To address these issues, Onerel [7] proposes to address joint entity and relation extraction in one module and one step, which is shown in figure 1. OneRel uses knowledge graph embedding methods to assign a single table for each relation using a score-based classifier and a relation-specific HORN labeling approach. In this study, we propose to assign a single table for every relation and use a multi-label classification loss [18]. The label of a token can belong to more than one relation. Therefore we can unify all relations in one table to save computation load.

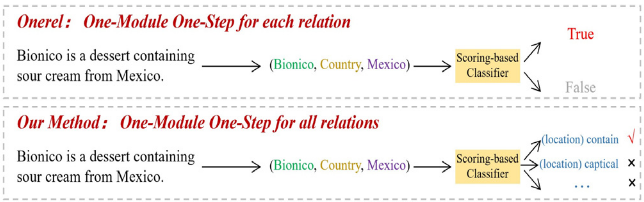

Figure 1. This figure demonstrates the classification paradigm of Onerel and our model.

While pre-train language models such as BERT [8] and its derivations (Deberta [9], Roberta [10]) have achieved excellent performance on many downstream NLP tasks jointly extracting entities and relations, it requires training with a large number of labeled data. Due to the expensive labeling of a joint entity and relation extraction task and different domain needs specific labeled data, it is important to improve the model performance on few-shot cases and domain adaption.

To address the issue of limited data, a promising area of research is meta-learning. One example of this is Model-Agnostic Meta-Learning (MAML [11]), which has shown excellent performance in diverse NLP tasks, including machine translation [12], relational classification [13], text summarization [14], and natural language understanding [15].

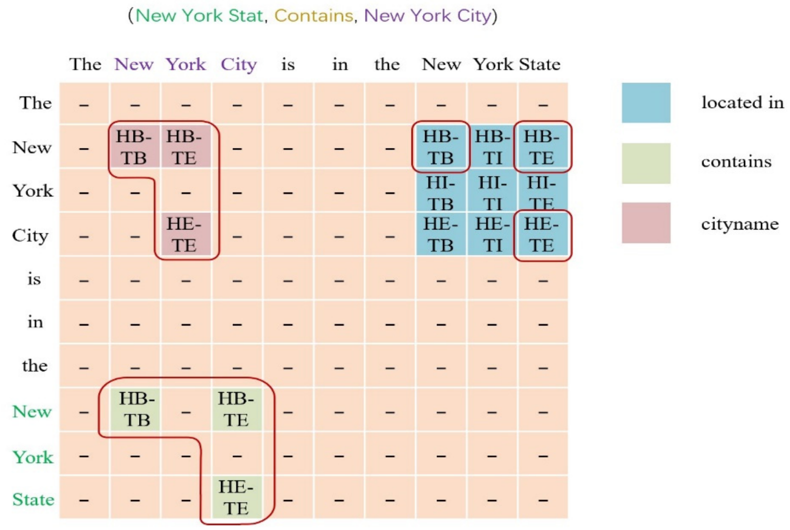

Figure 2. Examples of the tagging schema of Modified-Onerel. We predict all relations in one table using multi-label classification loss, one unit in the table can belong to more than one label.

MAML is based on the premise that tasks with similarities share common knowledge, and the algorithm converts this shared information from the source tasks to the form of weight initialization. The purpose is to train the learning ability of model and improve its generalization. With this learned initialization, the model can then learn new tasks in low-resource settings. Inspired by MTL-ABS [14], which uses meta-learning and pretraining of language models for low-resource abstractive summarization, we apply this technique to the joint entity and related tasks. In detail, we pre-train the bert model on a large-scale open-domain corpus, and then perform meta-learning on several source tasks, and finally fine-tune the model with 10 or 100 samples on the target task. Following MTL-ABS [14], we adopt adapters [16] on bert for few-shot learning to avoid overfitting for a large model with many parameters [17].

In this study, we introduce MTL-JER, a Meta-Transfer Learning method for Joint Entity and Relation Extraction with low-resource conditions. Firstly, we propose a unified framework because of Onerel, which can address all relations in just one table. Then we adopt meta-learning with adapters to perform few-shot learning, which improves the model generalization ability and domain adaptation ability. Experiments on nine benchmark joint entity and relation extraction datasets indicate that MTL-JER achieves optimal achievement with low-resource compared to precious work. We further added analysis and ablation study to prove that: 1) our unified framework has fewer parameters and achieves better results than Onerel. 2) Meta-transfer learning with adapters is effective in few-shot learning cases.

II. PRELIMINARIES

A. Onerel

For input sentences, Onerel first employs a pre-trained BERT [8] as a sentence encoder to generate embeddings for every token e_i:

$$\{\mathrm{e}_1, \mathrm{e}_2, \ldots, \mathrm{e}_L\} = BERT(\{x_1, x_2, \ldots, x_L\}) \tag{1}$$

Where x_i represents the input token, L represents the length of the sentence. Then Onerel uses a scoring classifier from HOLE [19]:

$$f(h,t) = r^T(s * o) \tag{2}$$

Where s and o are a representation of the subject and object entity. * means circular correlation, which can model the underlying relationships between two entities. The * operator is defined as a non-linear combination projection.

$$s * o = \mathrm{ReLU}\ (W[s;o]^T) + \mathrm{b} \tag{3}$$

Where $W \in R^{de\times 2d}$, d_e is the dimension of entity pair representations. [;] denotes the operation of concatenation.

Then we randomly initialized relation representation $R^T \in R^{de\times 4K}$, and the evaluation function of the method is:

$$v_{(w_i, r_k, w_j)} = R^T \mathrm{ReLU}\ (W[s;o]^T) + \mathrm{b}\Big) \tag{4}$$

Where K represents the total amount of relations, 4 is the number of classification tags. Onrel uses Relation Specific Horns Tagging to extract entity pairs in a table: Head Begin-Tail End(HBTE), Head Begin-Tail Begin(HB-TB), Head End-Tail End(HE-TE), and other(-), head and tail refers to subject and object separately.

B. Adapters

Recent research has demonstrated that fine-tuning all the parameters of pre-trained language models (PLMs) with extensive parameters on low-resource datasets can result in overfitting ([17], [20]). To address this issue, Pfeiffer et al. [16] introduces small task-specific modules called adapters into the PLM layers and trains only the additional adaptors and layer normalization while keeping the other PLM parameters fixed.

Each adapter in a layer is represented as A(.) ∈ RH, comprised of a down-projection, D(.) ∈ RH×B, an activation function, and an up-projection U(.) ∊ RB×H, where H represents the dimension of the input hidden states x, and B represents the adapter bottleneck hidden size. The formula is:

$$A(x)=U(\mathrm{GeLU}\,(D(x)))+x \tag{5}$$

III. METHODOLOGY

We would introduce the modified Onrel model in this part and then how to apply meta-learning for low-resource joint entities and related tasks.

A. Modified-Onerel

For Onerel, for each relation, it constructs a table for each relation, and the model predicts a set of HB-TE, HE-TE, and HB-TB to extract a pair entity. But such a method still needs to construct an N table, which has a great computation amount, and most of the spans in the table are negative samples. To address these issues, we propose Modified-Onerel, using multi-label classification loss [18], such that an entity pair can belong to more than one relation. Therefore we can use one table to represent all relations, which is shown in figure II. In detail, for a token $e_{i,j}$ in a table, the possible label for it is E+K, where E belongs to an entity or not, which is equal to 2. K is the number of relation types.

Such a design has the following merits: 1) unify all relations in one table, which can reduce relation redundancy and computation. 2) Being able to address overlapped relations and entities due to multi-label classification. 3) This design can be further expanded to predict the type of entities by adding the entity type into the label set.

The multi-label classification loss is defined as:

$$\log\left(\sum_{i\in\Omega_{\mathrm{neg}},j\in\Omega_{pos}} e^{s_i-s_j}+\sum_{i\in\Omega_{\mathrm{neg}}} e^{s_i-s_0}+\sum_{j\in\Omega_{pos}} e^{s_0-s_j}\right)$$
$$=\log\left(e^{s_0}+\sum_{i\in\Omega_{neg}} e^{s_i}\right)+\log\left(e^{-s_0}+\sum_{j\in\Omega_{pos}} e^{-s_j}\right) \tag{6}$$

Where s_0 is the threshold, when the probability of one label is above the threshold, then the prediction belongs to the label.

Here we set the threshold as 0, and the formula can be set as:

$$=\log\left(1+\sum_{i\in\Omega_{neg}} e^{s_i}\right)+\log\left(1+\sum_{j\in\Omega_{pos}} e^{-s_j}\right) \tag{7}$$

Based on equation 4 and Modified-Onerel, we only use one relation embedding $R^T \in R^{de\times 4K}$ for all relations. The prediction probability of the score is:

$$P\left(y_{(w_i,r,w_j)}\mid S\right)=\mathrm{Softmax}\left(v_{(w_i,r,w_j)}\right) \tag{8}$$

The loss function of the modified-OneRel is :

$$L_{\text{triple}}=-\frac{1}{L\times L}\times\sum_{i=1}^{L}\sum_{j=1}^{L}\log P\left(y_{(w_i,r,w_j)}=g_{(w_i,r,w_j)}\mid S\right) \tag{9}$$

B. Meta-Transfer Learning for Low-Resource Joint Entity and Relation Extraction

Low-Resource Joint Entity and Relation Extraction involves training a model on source corpus data to acquire experience, followed by fine-tuning a few samples of target corpus data and evaluating the model using relevant metrics. To perform low-resource learning, we employ meta-transfer learning as it provides a better initialization for the model, allowing it to learn more effectively and efficiently on new data.

Using adaptors in the model offers several advantages: a) fine-tuning all PLM parameters with multitudinous parameters is not effective sampling and may be unstable under low resource settings [17]; adaptors allow for more efficient fine-tuning with retaining PLM parameters fixed, b) adaptors reduce both in-memory and CUDA memory requirements by having fewer trainable parameters, and c) they also enhance the stability of PLMs, making them an appropriate method for few-shot fine-tuning [21].

Following the method used in MTL-ABS [14], we use an adapter-enhanced model and apply meta-transfer learning to enable fast adaptation to new corpora. We use a pretraining corpus (C^{pre}), a set of source corpora $\{C_j^{\mathrm{src}}\}$, and a target corpus (C^{tgt}) with limited labeled examples to make better the model on the target corpus. By leveraging both C^{pre} and the source corpora $\{C_j^{\mathrm{src}}\}$, we aim to enhance the performance of the low-resource target corpus.

Our models contain Modified-Onerel θ and adapters ϕ. First, we train the Modified-Onrel model θ $\{C_j^{\mathrm{src}}\}$using a large-scale open-domain dataset to train the ability of model on the joint entity and relation extraction task. Then we perform load the trained Modified-Onrel model, fix its parameters, and add adapters, then perform meta-learning $\{C_j^{\mathrm{src}}\}$with multi-domain, and finally, further train the model of meta-learned on 10 or 100 samples on C^{tgt}. The training of meta-learning is as follows:

First, we sampled from source corpora $\{C_j^{\mathrm{src}}\}$ without replacement to obtain a set of tasks $\{\mathcal{T}_i\}_{i=1}^{\mathcal{M}}$ as a meta-dataset. For one task, Ti contains a support set $Q_i^{tr}=\left\{\left(\mathbf{x}_i^k,\mathbf{y}_i^k\right)\right\}_{k=1}^{K}$ and a query set $Q_i^{te}=\left\{\left(\mathbf{x}_i^{*k},\mathbf{y}_i^{*k}\right)\right\}_{k=1}^{K}$. Support and query set correspond to the train and test set in task data. Meta-learning is a bi-level optimization process containing inter loop and an outer loop. We initialize the model with parameters θ and ϕ and only update ϕ. For inner loop optimization, on each task T_i. Firstly, we train the model on the support set and update the model on the loss of query set $Q_i^{te}=\left\{\left(\mathbf{x}_i^{*k},\mathbf{y}_i^{*k}\right)\right\}_{k=1}^{K}$:

$$\phi_i'=\phi-\alpha\nabla_\phi L_{T_i}\left(f_{\theta,\phi}\right) \tag{10}$$

The adapters parameters ϕ are trained to optimize the performance of $f_{\theta,\phi_i'}$, and then perform several gradient steps with few samples from tasks sampled from total tasks p(T). This can be realized by using the meta-objective:

$$\min_{\phi} \sum_{\mathcal{T}_i \sim p(\mathcal{T})} \mathcal{L}_{\mathcal{T}_i}\left(f_{\theta,\phi_i'}\right) = \sum_{\mathcal{T}_i \sim p(\mathcal{T})} \mathcal{L}_{\mathcal{T}_i}\left(f_{\theta,\phi-\alpha\nabla_\phi \mathcal{L}_{\mathcal{T}_i}(f_{\theta,\phi})}\right) \tag{11}$$

For the outer loop, the optimization of the meta-objective is performed over tasks by gradient descent by making updates to ϕ:

$$\phi \leftarrow \phi - \epsilon \nabla_\theta \sum_{\mathcal{T}_i \sim p(\mathcal{T})} \mathcal{L}_{\mathcal{T}_i}\left(f_{\theta,\phi_i'}\right)$$

Where ϵ represents the meta step size parameter.

The goal of meta-learning is to seek out meta parameters ϕ∗ that can efficiently learn from a little support set and extend it to query sets in the same domain. This is important because, in each task, only a limited number of examples are available in the support set. The "learn to learn" paradigm makes meta-learning particularly effective for few-shot learning scenarios.

TABLE I. TEST F1-SCORE(%)ON JER DATASET

Method	ACE2004		ACE2005		SciERC		NYT		WEBNLG	
	partial	*exact*	*partial*	*exact*	*partial*	*exact*	*partial*	*exact*	*partial*	*exact*
10 trained samples										
FTL	11.52	10.40	12.79	12.03	6.39	3.75	19.69	17.30	20.16	18.27
ATL	14.89	14.08	15.30	14.28	7.24	5.13	26.45	25.28	24.59	23.05
DA	16.09	12.78	15.24	14.80	10.37	7.55	25.74	22.46	28.53	25.25
MTL-JER	**23.53**	**21.65**	**21.53**	**21.48**	**14.53**	**12.33**	**32.53**	**31.56**	**37.42**	**35.53**
100 trained samples										
FTL	15.53	12.84	16.73	15.67	9.43	5.53	30.52	29.73	32.93	32.77
ATL	16.43	14.64	18.46	17.30	12.52	10.09	34.54	33.80	35.65	33.53
DA	25.89	25.00	24.03	22.19	15.53	13.97	44.52	44.24	47.28	45.53
MTL-JER	**36.74**	**35.67**	**31.58**	**30.76**	**20.07**	**17.98**	**52.65**	**50.90**	**55.17**	**53.84**

IV. EXPERIMENTS

A. Implementation Details

In this work, our implementation leverages the Huggingface Transformers Library [22]. We perform pretraining with pre-train corpus using FewRel dataset [23]. We use 5 datasets evaluation; for the dataset chosen as the target corpus, we use the rest of the 4 datasets for meta-learning and select 10 or 100 samples from the target corpus for fine-tuning. For meta-learning, the meta-batch size is set to 2, and the batch size for each task is 3. The total number of support sets and query sets is set as 8,000 and 4,000 for each task. We conduct hyper-parameter searches, including batch size, learning rate, and warm-up ratio. The best setting is batch size 64, learning rate 105, and warm-up ratio 0.1. We use AdamW as an optimizer and linear scheduler for warm-up. We trained the model for 25 epochs most, and using an early stop to avoid overfitting, the patience for an early stop is 8. We perform validation every 50 steps during training and save the best performing model on the development dataset. To eliminate the effects of random seeds, we set five sets of random seeds: 50, 100, 200, 300, and 500, to run 5 experiments, and the results are the average of the five experiments.

Our experiments were carried out on a single NVIDIA 3080 GPU using PyTorch, with bert-base-uncased as the chosen pre-trained model.

B. Dataset

FewRel[22] is composed of 70,000 sentences about 100 relationships.

We use the SciERC [24] dataset that comes from 500 abstracts of AI articles and scientific terms designed for the construction of scientific knowledge graphs. The preprocessing steps are the same as those in a previous study [25].

The **ACE2004**[2] and **ACE2005**[3] datasets are drawn from various sources, consist of news articles and online forums.

The **NYT [26],** the dataset used for distantly supervised relation extraction, consists of 24 relation types and includes a test group of 5000 sentences, a validation group of 5000 sentences, and a training group of 56195 sentences. Note that this dataset contains overlapping relations.

WebNLG [27] contains 246 relation types, 5019 sentences for training, 703 sentences for testing, and 500 sentences for validation. This dataset contains overlapped relations.

TABLE II. TEST F1-SCORE(%)ON JER DATASET

Method	NYT			WEBNLG		
	precision	*recall*	*f1*	*precision*	*recall*	*f1*
Onerel	94.2	92.6	92.9	91.8	90.3	91.0
Modified-Onerel	**94.8**	**93.1**	**93.9**	**91.9**	**90.8**	**91.4**

C. Baselines

We compare our model with fine-tuning on full dataset and data augmentation methods for few-shot learning.

Fine-tune transfer learning(FTL) After pretraining on a pre-train corpus, we fine-tuned the BERT model with either 10 or 100 labeled target examples.

Adapters transfer learning(ATL) After pretraining on a pre-train corpus, we fine-tune the adapter modules with 10 or 100 labeled target examples.

Data Augmentation[28](DA) After pretraining on a pre-train corpus, we use data augmentation strategies in Li et al. [28], such as entity and context permutation, to generate ten times samples for training, and we fine-tune the adapter modules with the augmented examples.

D. Experimental Results and Analysis

Experiments results are displayed in Table 1, where the evaluation metric used is the micro F1-score. The evaluation method adopted is the exact and partial match, as described in Liu et al .[29]. In the case of exact matching, a predicate triple is considered right only when the relationship and the full nouns of both head and tail entities are right. In partial matching, the predicate triple is only considered correct if the relation and the final word of the head and tail entity names are both right.

We show that MTL-JER outperforms prior; however, when fine-tuning with low-resource data, the model is prone to overfitting, leading to poor performance on the test set, especially when trained with only 10 labeled data points. Trained with adapters reduces the parameters to be updated, but still exists the overfitting problem. The data augmentation method has limited boosting effects because the augmented data is similar to the original data and, thus, is not of high quality. In contrast, MTL-JER can achieve remarkable results even when trained on 100 labels. This is due to the stage of meta-learning; the models learn to learn each task instead of simply 'remembering' the trained data, which results in overfitting.

TABLE III. TEST F1-SCORE(%)ON JER DATASET

Method	NYT			WEBNLG		
	train	*dev*	*test*	*train*	*dev*	*test*
w/o. adapter	73.53	20.56	17.30	79.43	23.53	18.27
w/. adapter	49.42	22.78	**25.28**	47.9	25.58	**23.05**

E. Ablation Study

1) Effectiveness of Modified-Onrel

Finally, to verify the effectiveness of modification on Onerel, we perform fine-tuning experiments on both the NYT and WebNLG datasets. As we can see in table II, Modified-Onrel performs better than Onerel, indicating that by unifying all relationships, the model can learn mutual relationships between relationships and reduce the quantity of negative samples to avoid label imbalance.

2) Preventing Overfitting with Adapters

In this part, we would validate the effectiveness of the adapters module in preventing overfitting. Training on low-resource tends to have more gradient steps on large parameters to ensure fit the training data and causes overfitting. Only training parameters of adapter parameters can alleviate this issue. Table III shows the results of adopting adapters with 10 train sets, as shown the full model fine-tuning has a high score on the train set but low on the test set, this shows that fine-tuning of the whole model is easy to overfit, resulting in worse generalization ability.

V. CONCLUSION

This paper presents that MTL-JER, which can process joint entity and relation extraction tasks in a unified and parameter-efficient way. We raise the problem in the Onerel model and put forward solutions. We further apply the model to low-resource scenarios, using meta-transfer learning to conduct few-shot learning. We apply adapters structure on the pre-trained language model to prevent the few-shot learning from overfitting. By performing pretraining and meta-learning, our model can achieve remarkable results on a few training data. Experiments of five datasets show that our model is effective compared with previous work.

ACKNOWLEDGEMENT

The work is supported by the Science and Technology on Complex Electronic System Simulation Laboratory Fund Project(Item No:6142401004032106)

REFERENCES

[1] Zelenko, D.; Aone, C.; and Richardella, A. 2003. Kernel Methods for Relation Extraction. J. Mach. Learn. Res., 3: 1083–1106.

[2] Chan, Y. S.; and Roth, D. 2011. Exploiting SyntacticoSemantic Structures for Relation Extraction. In Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies, 551– 560. Portland, Oregon, USA: Association for Computational Linguistics.

[3] Li, Q.; and Ji, H. 2014. Incremental Joint Extraction of Entity Mentions and Relations. In Proceedings of the 52nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), 402–412. Baltimore, Maryland: Association for Computational Linguistics.

[4] Bowen Yu, Zhenyu Zhang, Xiaobo Shu, Yubin Wang, Tingwen Liu, Bin Wang, and Sujian Li. 2020. Joint extraction of entities and relations based on a novel decomposition strategy. In Proceedings of ECAI.

[5] Wei Z, Su J, Wang Y, et al. A novel cascade binary tagging framework for relational triple extraction[J]. arXiv preprint arXiv:1909.03227, 2019.

[6] Wang Y, Yu B, Zhang Y, et al. TPLinker: Single-stage joint extraction of entities and relations through token pair linking[J]. arXiv preprint arXiv:2010.13415, 2020.

[7] Shang Y M, Huang H, Mao X. Onerel: Joint entity and relation extraction with one module in one step[C]//Proceedings of the AAAI Conference on Artificial Intelligence. 2022, 36(10): 11285-11293.

[8] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019.BERT: Pretraining of deep bidirectional transformers for language understanding[J]. In Proceedings of the 2019 Conference of the North

American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers). 2019: 4171–4186.

[9] He P, Liu X, Gao J, et al. Deberta: Decodingenhanced bert with disentangled attention[J]. arXiv preprint arXiv:2006.03654, 2020.

[10] Liu Y, Ott M, Goyal N, et al. Roberta: A robustly optimized bert pretraining approach[J]. arXiv preprint arXiv:1907.11692, 2019.

[11] Finn, C.; Abbeel, P.; and Levine, S. 2017. ModelAgnostic Meta-Learning for Fast Adaptation of Deep Networks. In Precup, D.; and Teh, Y. W., eds., Proceedings of the 34th International Conference on Machine Learning, volume 70 of Proceedings of Machine Learning Research, 1126–1135.

[12] Li, R.; Wang, X.; and Yu, H. 2020. MetaMT is a Meta-Learning Method Leveraging Multiple Domain Data for Low Resource Machine Translation. Proceedings of the AAAI Conference on Artificial Intelligence 34(05): 8245–8252.

[13] Obamuyide, A.; and Vlachos, A. 2019. Model-Agnostic Meta-Learning for Relation Classification with Limited Supervision. In Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics, 5873–5879.

[14] Chen Y S, Shuai H H. Meta-Transfer Learning for LowResource Abstractive Summarization[C]//Proceedings of the AAAI Conference on Artificial Intelligence. 2021, 35(14): 12692-12700.

[15] Dou, Z.-Y.; Yu, K.; and Anastasopoulos, A. 2019. Investigating Meta-Learning Algorithms for Low-Resource Natural Language Understanding Tasks. In Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLPIJCNLP), 1192–1197.

[16] Jonas Pfeiffer, Andreas Ruckl¨ e, Clifton Poth, Aishwarya' Kamath, Ivan Vulic, Sebastian Ruder, Kyunghyun' Cho, and Iryna Gurevych. 2020. Adapterhub: A framework for adapting transformers. In EMNLP: System Demonstrations.

[17] Zintgraf, L.; Shiarli, K.; Kurin, V.; Hofmann, K.; and Whiteson, S. 2019. Fast Context Adaptation via MetaLearning. In Chaudhuri, K.; and Salakhutdinov, R., eds., Proceedings of the 36th International Conference on Machine Learning, volume 97 of Proceedings of Machine Learning Research, 7693–7702.

[18] Su J, Murtadha A, Pan S, et al. Global Pointer: Novel Efficient Span-based Approach for Named Entity Recognition[J]. arXiv preprint arXiv:2208.03054, 2022.

[19] Nickel, M.; Rosasco, L.; and Poggio, T. 2016. Holographic embeddings of knowledge graphs. In Proceedings of the AAAI Conference on Artificial Intelligence, volume 30.

[20] Jesse Dodge, Gabriel Ilharco, Roy Schwartz, Ali Farhadi, Hannaneh Hajishirzi, and Noah Smith. 2020. Fine-tuning pre-trained language models: Weight initializations, data orders, and early stopping. arXiv preprint arXiv:2002.06305.

[21] Mahabadi R K, Zettlemoyer L, Henderson J, et al. Perfect: Prompt-free and efficient few-shot learning with language models[J]. arXiv preprint arXiv:2204.01172, 2022.

[22] Wolf T, Debut L, Sanh V, et al. Transformers: State-of-the-art natural language processing[C]. Proceedings of the 2020 conference on empirical methods in natural language processing: system demonstrations. 2020: 38-45.

[23] Han X, Zhu H, Yu P, et al. Fewrel: A large-scale supervised few-shot relation classification dataset with state-of-the-art evaluation[J]. arXiv preprint arXiv:1810.10147, 2018.

[24] Yi Luan, Luheng He, Mari Ostendorf, and Hannaneh Hajishirzi. 2018. Multi-task identification of entities, relations, and coreference for scientific knowledge graph construction. In Empirical Methods in Natural Language Processing (EMNLP), pages 3219–3232.

[25] Zhong Z, Chen D. A frustratingly easy approach for entity and relation extraction[J]. arXiv preprint arXiv:2010.12812, 2020.

[26] S. Riedel, L. Yao, and A. McCallum, "Modeling relations and their mentions without labeled text," in ECML, 2010, pp. 148–163.

[27] C. Gardent, A. Shimorina, S. Narayan, and L. PerezBeltrachini, "Creating training corpora for NLG micro planners," in ACL, 2017, pp. 179–188.

[28] Li L, Chen X, Ye H, et al. On robustness and bias analysis of bert-based relation extraction[C]//China Conference on Knowledge Graph and Semantic Computing. Springer, Singapore, 2021: 43-59.

[29] J. Liu, S. Chen, B. Wang, J. Zhang, N. Li, and T.Xu, "Attention as relation: Learning supervised multi-head self-attention for relation extraction," in IJCAI, 2020, pp. 3787–3793.